

Amir Rafe, Ph.D.

Postdoctoral Researcher

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Profile

Transportation safety researcher with a Ph.D. in Civil and Environmental Engineering and an active portfolio of federally sponsored safety research across NCHRP, BTSCRP, FMCSA, and the AAA Foundation. Specializes in crash and injury severity modeling under unobserved heterogeneity, safety performance functions and crash modification factors, vulnerable road user and nighttime safety, and causal AI for proactive safety decision support. Fifteen years of programming experience building reproducible pipelines that integrate crash, roadway, telematics, exposure, and spatial data, with a record in proposal development, sponsor interaction, and supervision of junior researchers.

Areas of Expertise

Research	Traffic safety and operations; vulnerable road users; freight and commercial-vehicle risk; causal and trustworthy AI; intelligent transportation systems; evacuation modeling; infrastructure resilience.
Safety Analytics	Highway Safety Manual; safety performance functions and crash modification factors; network screening; MUTCD and traffic control devices; Safe System Approach; FARS, CRSS, and state crash databases; naturalistic driving and telematics data; lighting and visibility conditions in crash modeling.
Software	Python; R; MATLAB; SQL (SQL Server, PostgreSQL, BigQuery); ArcGIS/QGIS and scripted geospatial analysis; PTV Vissim/Viswalk; SUMO; MATSim; AutoCAD Civil 3D; Highway Capacity Software.

Education

- 2022–2025 **Ph.D. in Civil & Environmental Engineering**, *Utah State University*, Transportation Engineering specialization, GPA: 4.0
Dissertation: [EvacuAIDi: An AI-Driven, Causal-Informed Framework for Probabilistic and Disability-Inclusive Evacuation Guidance](#)
- 2010–2013 **M.S. in Civil & Environmental Engineering**, *Azad University*, Transportation Engineering specialization, GPA: 4.0
Thesis: Pedestrian Dynamics Modeling Using the Social Force Model: A Heterogeneous Approach
- 2005–2009 **B.S. in Civil Engineering**, *Azad University*, GPA: 4.0

Academic and Professional Appointments

- 2025–Present **Postdoctoral Researcher**, [AI in Transportation \(AIT\) Lab](#), Texas State University, Ingram School of Engineering
- 2022–2025 **Graduate Research Assistant**, [Singleton Transportation Lab](#), Utah State University, Department of Civil & Environmental Engineering

2012–2022 **Algorithm Engineer**, *Ramona T&I and IUST Research Institutes*
Transportation modeling, simulation, software development, and applied traffic-safety studies.

Publications

Peer-Reviewed Journal Articles

- 2026 **Rafe, A.**, & Das, S. “Do Aviation Hazard Rankings Overlook Conservation-Concern Birds in U.S. Wildlife Strikes?” *Transportation Research Part D: Transport and Environment*, 161, 105602. <https://doi.org/10.1016/j.trd.2026.105602>.
- Rafe, A.**, Singleton, P. A., & Christensen, K. M. “Experimental Analysis and Disability-Aware Social Force Modeling of Heterogeneous Evacuation with Counterfactual Route Guidance.” *Physica A: Statistical Mechanics and its Applications*, 131992. <https://doi.org/10.1016/j.physa.2026.131992>.
- GBD 2023 Road Injuries Collaborators (**Rafe, A.**, named collaborator). “Global, regional, and national burden of road injuries 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023.” *The Lancet Public Health*. [https://doi.org/10.1016/S2468-2667\(26\)00141-6](https://doi.org/10.1016/S2468-2667(26)00141-6).
- Barua, S., Starewich, M., Chowdhury, T. I., **Rafe, A.**, & Das, S. “Decoding Pedestrian Severity at Crosswalks using Hybrid Clustering and Random Parameter Models.” *Scientific Reports*. <https://doi.org/10.1038/s41598-026-56530-5>.
- 2025 Somvanshi, S., Islam, M. M., **Rafe, A.**, Tusti, A. G., Chakraborty, A., Baitullah, A., Chowdhury, T. I., Alnawmasi, N., Dutta, A., & Das, S. “Bridging the Black Box: A Survey on Mechanistic Interpretability in AI.” *ACM Computing Surveys*. <https://doi.org/10.1145/3787104>.
- Rafe, A.**, Singleton, P. A., Boyer, S., & Mekker, M. “Pedestrian Crossing Behaviors at Signalized Intersections in Utah: Factors Affecting Spatial and Temporal Violations.” *International Journal of Transportation Science and Technology*. <https://doi.org/10.1016/j.ijtst.2024.12.005>.
- 2024 **Rafe, A.**, Arman, M. A., & Singleton, P. A. “A Comparative Study Using Generalized Ordered Probit, Stacking Ensemble, and TabNet: Application to Determinants of Pedestrian Crash Severity.” *Data Science for Transportation*, 6(2), 13. <https://doi.org/10.1007/s42421-024-00098-x>.
- Rafe, A.**, & Singleton, P. A. “Imputing Time Series Pedestrian Volume Data With Consideration of Epidemiological-Environmental Variables.” *Transportation Research Record*. <https://doi.org/10.1177/03611981241240758>.
- Rafe, A.**, & Singleton, P. A. “Exploring the Complexity of Pedestrian Dynamics: Impact of Societal Behaviors and Personal Attributes in Urban Environments.” *Transportation Research Record*. <https://doi.org/10.1177/03611981241260707>.

Conference Proceedings and Presentations

- 2025 **Rafe, A.**, Lawrence, P. J., Lovreglio, R., Spearpoint, M., & Singleton, P. A. “Enhancing Occupant Evacuation Simulation Using LLMs and Retrieval-Augmented Generation.” *ICTD 2025*. <https://doi.org/10.1061/9780784486191.034>.
- Afand, K., **Rafe, A.**, Khademi, N., Mazloun, S., Ahmadi, A., Zabihpour, A., & Singleton, P. A. “Virtual Reality Analysis of Leadership Dynamics and Pedestrian Route Choice During Mass Gathering Evacuations.” *ICTD 2025*. <https://doi.org/10.1061/9780784486207.004>.

- Mazloun, S., Khademi, N., Ahmadi, A., Afand, K., Zabihpour, A., **Rafe, A.**, & Singleton, P. A. "Exploring Pedestrian Evacuation Behavior Using a Virtual Reality Environment: An Eye-Tracking Study." *ICTD 2025*. <https://doi.org/10.1061/9780784486207.035>.
- 2024 **Rafe, A.**, & Singleton, P. A. "Exploring the Determinants of Pedestrian Crash Severity Using an AutoML Approach." *ICTD 2024*, 442–455. <https://doi.org/10.1061/9780784485514.039>.
- 2023 **Rafe, A.**, Kretz, T., & Singleton, P. A. "Importance of Social Behaviors on Pedestrian Dynamics: A Case Study of Islamic Clothing in Iran." *TRB Annual Meeting*.
- 2019 Arman, M. A., **Rafe, A.**, & Kretz, T. "Applied Hybrid Binary Mixed Logit to Investigate Pedestrian Crossing Safety at Midblock and Unsignalized Intersection." arXiv. <https://doi.org/10.48550/arXiv.1905.09403>.
- 2015 Arman, M. A., **Rafe, A.**, & Kretz, T. "Pedestrian Gap Acceptance Behavior, A Case Study: Tehran." *TRB Annual Meeting*.
- 2014 **Rafe, A.**, & Karimi, M. "Calibrating Social Force Model Based on Design Experiments Method." *13th International Conference on Traffic and Transportation Engineering*.
- 2014 **Rafe, A.**, & Khavarzade, R. "Investigation of Pedestrian Gap Acceptance Behavior at Crosswalk." *13th International Conference on Traffic and Transportation Engineering*.
- 2014 **Rafe, A.**, Shariat, A., Kalantari, N., Arman, M. A., & Yazdanpanah. "Pedestrian Simulation with Viswalk at Various Walking Facilities." *13th International Conference on Traffic and Transportation Engineering*.

Manuscripts Under Review (selected)

- 2026 **Rafe, A.**, & Das, S. "From Deidentified Crash Narratives to Safety Intelligence Systems for Vulnerable Road Users." Submitted to *Hawaii International Conference on System Sciences (HICSS)*.
- Rafe, A.**, & Das, S. "Marginal Validity Is Not Enough for Equitable Conformal Prediction of Public AI Attitudes." Submitted to *Hawaii International Conference on System Sciences (HICSS)*.
- Rafe, A.**, & Das, S. "Differential Reporting and Recovered Behavioural Mechanisms in Electric Scooter Injury Surveillance Under a Safe Systems Lens." Submitted to *Safety Science*.
- Yadav, R., **Rafe, A.**, Chhetri, G., & Das, S. "Do Not Prune the Crash: Criticality-Aware Compute Allocation with Conformal Rare-Event Recall Control for Streaming Transportation Video-VLMs." Submitted to *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.
- Rafe, A.**, & Das, S. "Community Driving-Safety Deterioration as a Push Factor for Public Endorsement of AI Driving Capability." Submitted to *Accident Analysis & Prevention*. <https://doi.org/10.48550/arXiv.2604.04775>.
- Rafe, A.**, Baitullah, A., & Das, S. "Latent Profiles of AI Risk Perception and Their Differential Association with Community Driving Safety Concerns: A Person-Centered Analysis." Submitted to *Transportation Research Part F: Traffic Psychology and Behaviour*. <https://doi.org/10.48550/arXiv.2604.04849>.
- Rafe, A.**, & Das, S. "Who Bears the Risk When Generative AI Enters Transport? A Distributional Sociotechnical Audit of Algorithmic Equity, Synthetic-Data Validity, and Public Trust." Submitted to *Transportation Research Part A: Policy and Practice*.
- Rafe, A.**, & Das, S. "Heterogeneous Ordinal Structure Learning with Bayesian Nonparametric Complexity Discovery." Submitted to *NeurIPS 2026*. <https://doi.org/10.48550/arXiv.2605.04191>.

- Rafe, A., & Das, S.** “Coupled-NeuralHP: Directional Temporal Coupling Between AI Innovation Exposure and Public Response.” Submitted to *NeurIPS 2026*. <https://doi.org/10.48550/arXiv.2605.04194>.
- Rafe, A., & Das, S.** “Socio-Conformal Calibration in Complex Survey Data: Marginal Validity Is Not Enough for Subgroup Reliability.” Submitted to *NeurIPS 2026*. <https://doi.org/10.48550/arXiv.2605.05562>.
- Rafe, A., & Das, S.** “From Aha Tokens to Convergent Insight: An Externalist Audit for Emergent LLM Self-Correction.” Submitted to *ACM AI Letters*.
- Rafe, A., & Das, S.** “The Double-Edged Algorithm: A Causal-Epistemic Framework for Generative AI and Researcher Creativity.” Submitted to *Technological Forecasting and Social Change*.
- Rafe, A., Brotee, S., Chhetri, G., Bellamkonda, V. S., & Das, S.** “From Control Theory to Reinforcement Learning and Language Models: A Survey on Safe Path Planning for Autonomous Systems.” Submitted to *ACM Computing Surveys*.
- 2025 **Rafe, A., & Das, S.** “Causal AI and Computational Epistemology: A Survey.” Submitted to *ACM Computing Surveys*. <https://dx.doi.org/10.2139/ssrn.6045495>.
- Chhetri, G., Brotee, S., Ansari, M. W., Bellamkonda, V. S., **Rafe, A.**, Alnawmasi, N., Dutta, A., & Das, S. “Machine Unlearning as an Auditable System Capability: An IMGC Evidence Synthesis.” Submitted to *ACM Transactions on Intelligent Systems and Technology (TIST)*.
- Brotee, S., Chhetri, G., Pollock, S. B. B., Bellamkonda, V. S., **Rafe, A.**, & Das, S. “A Survey on Joint Embedding Predictive Architectures and World Models.” Submitted to *ACM Computing Surveys*. <https://dx.doi.org/10.2139/ssrn.5772122>.
- Afand, K., **Rafe, A.**, Khademi, N., & Das, S. “Leadership and Pedestrian Route Choice in Outdoor Evacuation: Environmental and Behavioral Influences During Gatherings.” Submitted to *Transportation Research Record*.
- Baitullah, A., Chakraborty, A., **Rafe, A.**, & Das, S. “Who Sees Pedestrian and Bicyclist Safety as a Problem? Evidence from National Survey Data.” Submitted to *Accident Analysis & Prevention*.
- Somvanshi, S., Sheley, R., Shuvo, S. A., **Rafe, A.**, & Das, S. “Automated Vehicles in Low Visibility and Infrastructure-Limited Roadway Settings.” Submitted to *IEEE Transactions on Intelligent Transportation Systems*. <https://dx.doi.org/10.2139/ssrn.5387394>.
- Pollock, S. B., Starewich, M., Somvanshi, S., Javed, S. A., Barua, S., **Rafe, A.**, & Das, S. “Red-Light Running Involved Crash Severity: Heterogeneity, Temporal Stability, and Policy Implications.” Submitted to *Transport Policy*.
- Rafe, A.**, & Singleton, P. A. “Who Leads, Who Follows? A Review of Leadership and Guidance Dynamics in Evacuation.” Submitted to *Safety Science*. Included in doctoral thesis.
- Rafe, A.**, & Singleton, P. A. “Causal AI for Evacuation Modeling: A Modular Framework for Standards-Compliant Simulation Input Generation.” Submitted to *Transportation Research Part D: Transport and Environment*. Included in doctoral thesis.
- Rafe, A.**, & Singleton, P. A. “Evaluating AI-Driven Evacuation Guidance: A Causal-Intervention and Counterfactual Simulation Framework.” Submitted to *Transportation Research Part C: Emerging Technologies*. Included in doctoral thesis.
- Rafe, A.**, & Singleton, P. A. “A Hierarchical Bayesian Framework for AI-Guided Evacuations under Human Compliance Uncertainty.” Submitted to *Transportation Research Part C: Emerging Technologies*. Included in doctoral thesis.

Research Projects and Funding Experience

- 2026–Present FMCSA: *Telematics-Informed Crash Risk Modeling to Support Targeted Traffic Enforcement for CMV Safety*, Federal Motor Carrier Safety Administration.
- 2025–Present NCHRP 17-113: *Incorporating Safe System Approach into the NCHRP Report 500 Series*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine.
- NCHRP Synthesis 20-05 / Topic 56-05: *Traffic Analysis Practices for Non-Motorists*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine.
- NCHRP 17-134: *Center Line Buffer Areas for Safety: Implementation Guidelines and Tool*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine.
- BTSCR P BTS-39: *Toolkit for Reducing Substance-Impaired Driving for the Last Leg of the Journey*, Behavioral Traffic Safety Cooperative Research Program.
- BTSCR P BTS-42: *Guidelines for Authorizing, Implementing, and Operating Automated Traffic Enforcement Programs*, Behavioral Traffic Safety Cooperative Research Program.
- AAA Foundation: *Examining the Pre-Crash Circumstances Leading to Pedestrian Fatalities*.
- 2024 *AI-Powered Tools for Safe Evacuation of Individuals During Emergencies*, Center for Transformative Infrastructure Preservation and Sustainability (CTIPS).
- 2022–2024 *The UDOT Pedestrian Data Platform*, Utah State University.
- 2022 *Prediction of Traffic Congestion Patterns Using Deep Learning*, in cooperation with KU Leuven.
- 2021 *Fire Simulation and Crowd Emergency Evacuation Studies of Parsabad and Khoy Airports*, Iriana Consulting Engineers Co.
- 2017–2020 *Traffic Safety and Impact Studies*, Tehran, Rasht, Qom, and Isfahan, Iriana Consulting Engineers Co.
- 2013–2016 Transportation resilience, traffic guide-sign software, tunnel evacuation, and pedestrian simulation studies with IUST and industry partners.

Submitted or Under Review Proposals (selected)

- 2026 NSF: *AIMing: Neuro-Symbolic Verification for the Autonomous Energy-Mobility Nexus*, Artificial Intelligence, Formal Methods, and Mathematical Reasoning (AIMing).
- Amazon Research Awards: *RoadSafe Agents: Governed Multi-Agent AI for Proactive Road Safety Scoring*.
- Amazon Research Awards: *SafetyLens and SafeLM: Risk-Calibrated Multimodal Foundation Models for Auditable Traffic Crash Understanding*.
- 2025 NSF: *Sheaf-Theoretic Identifiability for Multi-Sensor Safety Systems*, Mathematical Foundations of Artificial Intelligence (MFAI).
- NSF: *PIVOT: Privacy-Preserving Infrastructure for Video Synthesis and Analysis in Open Transportation Safety*, Cyberinfrastructure for Sustained Scientific Innovation (CSSI).
- ARPA-I: *EvacuVerse: A National Causal Digital Twin for Disability-Inclusive Evacuation and Multi-Modal Re-Routing*.
- CREATE University Transportation Center: *Role of Emerging Transportation Technologies and Safety Initiatives in Mitigating Crashes in Coastal Communities*.

Intellectual Property and Innovation

2026 **CausalSafe**: System and method for multimodal neuro-symbolic causal inference, SCM-based counterfactual traffic-risk assessment, and proactive safety intervention using grounded large language models. *Invention disclosure submitted*, Texas State University.

PIVOT: System and method for privacy-preserving synthesis, co-reported privacy–utility certification, and federated sharing of transportation video, trajectories, and safety analytics. *Invention disclosure submitted*, Texas State University.

Sheaf-Theoretic Multi-Sensor Fusion: System and method for trustworthy multi-sensor fusion in roadway safety applications using sheaf-theoretic consistency modeling. *Invention disclosure submitted*, Texas State University.

EvacuAIDi: Accessibility-aware AI framework for probabilistic emergency evacuation guidance. *Innovation disclosure submitted*, Utah State University.

Teaching Experience

2025–2026 Guest Lecturer: *Traffic Engineering; AI in Civil Engineering; Roadway Infrastructure Safety*, Texas State University.

2022–2024 Guest Lecturer: *Transportation Planning; Transportation Data and Safety Analysis*, Utah State University.

2023 Curriculum Developer, *Empowering Teaching Excellence (ETE-10)*: Prepared asynchronous modules for undergraduate and graduate students in statistics, data science, and applied AI.

Mentoring

Texas State University **Ph.D.**: Sawgat Ahmed Shuvo. **M.S.**: Arka Chakraborty; Anika Baitullah; Sazzad Bin Bashar Polock; Tausif Islam Chowdhury. **B.S.**: Biplov Pandey; Irfan Sarwar Pranjol; Bidisha Shrestha. **B.S. Computer Science**: Gaurab Chhetri; Radha Yadav.

Lund University **M.S. co-supervisor**: Yousef Zoghi.

Academic and Professional Service

Conference Review Transportation Research Board (TRB) Annual Meeting Reviewer, 2022–Present; Annual Conference on Neural Information Processing Systems (NeurIPS), 2026; IEEE International Conference on Intelligent Transportation Systems (ITSC), 2024.

Journal Review Safety Science; Transportation Research Part A; Transportation Research Part D; Transportation Research Part F; Accident Analysis & Prevention; European Transport Research Review; International Journal of Transportation Science and Technology; Journal of Transport & Health; Journal of Transport Geography; Transportation Research Record; Transportation Research Interdisciplinary Perspectives.

Awards and Certifications

Award The BobCatalyst Innovation Accelerator Program (BIAP) Award (Texas State University); Graduate Enhancement Scholarship (Utah State University).

Teaching Empowering Teaching Excellence (ETE-10), Utah State University.

Technical Neural Networks and Deep Learning (DeepLearning.AI); Data Science Professional Certificate (IBM); Intro to Traffic Flow Modeling and Intelligent Transport Systems (EPFL); Assessment of Safety using PTV VISSIM (University of Waterloo); Operations Research using GAMS (Sharif University).

Technical Skills

Programming	Python; R; SQL (PostgreSQL, BigQuery); Apache Spark; Docker; C++.
Simulation	MATSim; SUMO; PTV Vissim/Viswalk; Pathfinder; MassMotion; PyroSim; Fire Dynamics Simulator (FDS).
Engineering Software	AutoCAD Civil 3D; Revit; SolidWorks; BIM and OpenBIM/IFC.
Cloud and GIS	Google Cloud Platform; AWS; QGIS; ArcGIS.

Professional Memberships

American Society of Civil Engineers (ASCE); Institute of Transportation Engineers (ITE); Global Burden of Disease (GBD) Collaborator Network, Institute for Health Metrics and Evaluation (IHME), University of Washington.

Languages

English, fluent; Persian, fluent; Spanish, moderate.

References

- **Dr. Subasish Das**, Associate Professor, Ingram School of Engineering, Texas State University. Email: subasish@txstate.edu; Phone: +1-512-245-1826.
- **Dr. Patrick Singleton**, Associate Professor, Department of Civil and Environmental Engineering, Utah State University. Email: patrick.singleton@usu.edu; Phone: +1-435-797-7109.
- **Dr. Ruggiero Lovreglio**, Professor in Digital Built Environment, School of Built Environment, Massey University. Email: R.Lovreglio@massey.ac.nz; Phone: +64-9-212-7114.